

AA4 – e-NFAs

1

A spontaneous transition from a state A to a state B in a ϵ -NFA means _____ and _____. 
(0/1 Ponto)

For the first missing snippet select one of the first two options and for the second snippet, from the remaining options, select the correct one(s).

- ☐ to process the current symbol of the input and ignore its value
- ☐ to not process the current symbol
- ☐ stay in A
- ☐ simply move to B
- ☐ build a new path moving to state B
- ☐ recursively build new paths depending on spontaneous transitions of B

2

The conversion of a ϵ -NFA a DFA will result in a DFA with _____ states.
(1/1 Ponto)

- ☐ *exactly $2 \times n$*
- ☐ *at most $2 \times n$*
- ☐ *exactly 2^n*
- ☐ *at most 2^n*
- ☐ *possibly more than 2^n*

3

We can use a DFA to represent a problem requesting a ϵ -NFA and, similarly, we can use a ϵ -NFA to represent when a NFA is requested.

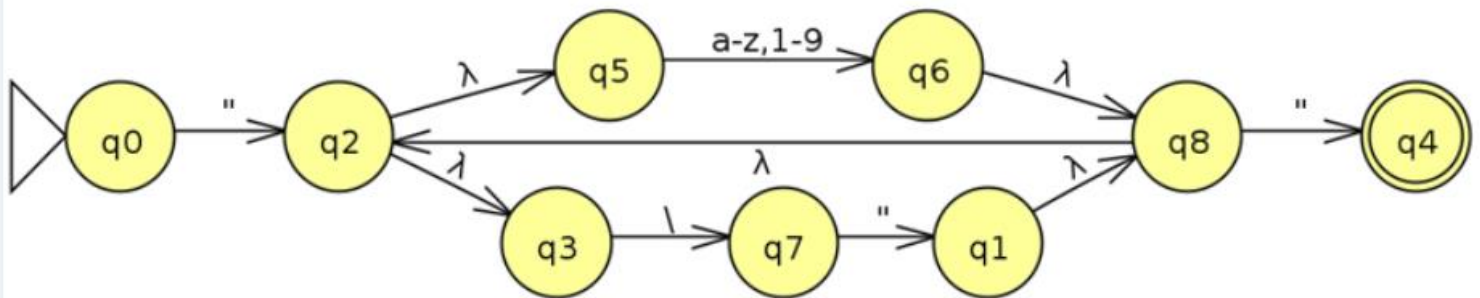
(0/1 Ponto)

- ☐ The sentence is true.
- ☐ The sentence is false.
- ☐ The sentence is true for the first part and false for the second part..
- ☐ The sentence is false for the first part and true for the second part.

4

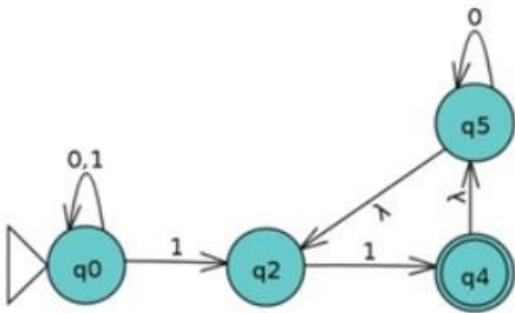
Which of the following strings are accepted by the automaton?

(1/1 Ponto)

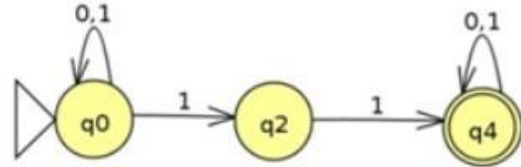


- ☐ "myfirsttest"
- ☐ "yet"another"test"
- ☐ "2more\"2go"
- ☐ "a\"*2"
- ☐ ""
- ☐ <empty string>

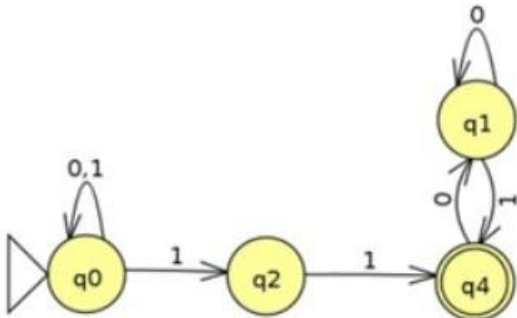
Which of the following automata represent the same language?
(0/1 Ponto)



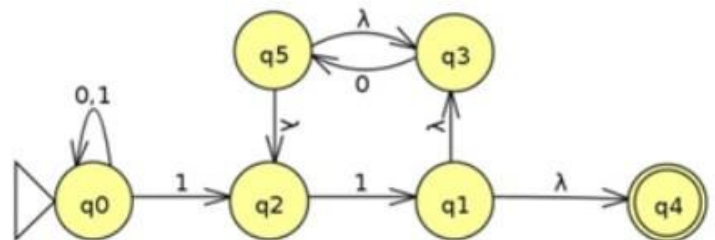
a)



b)



c)

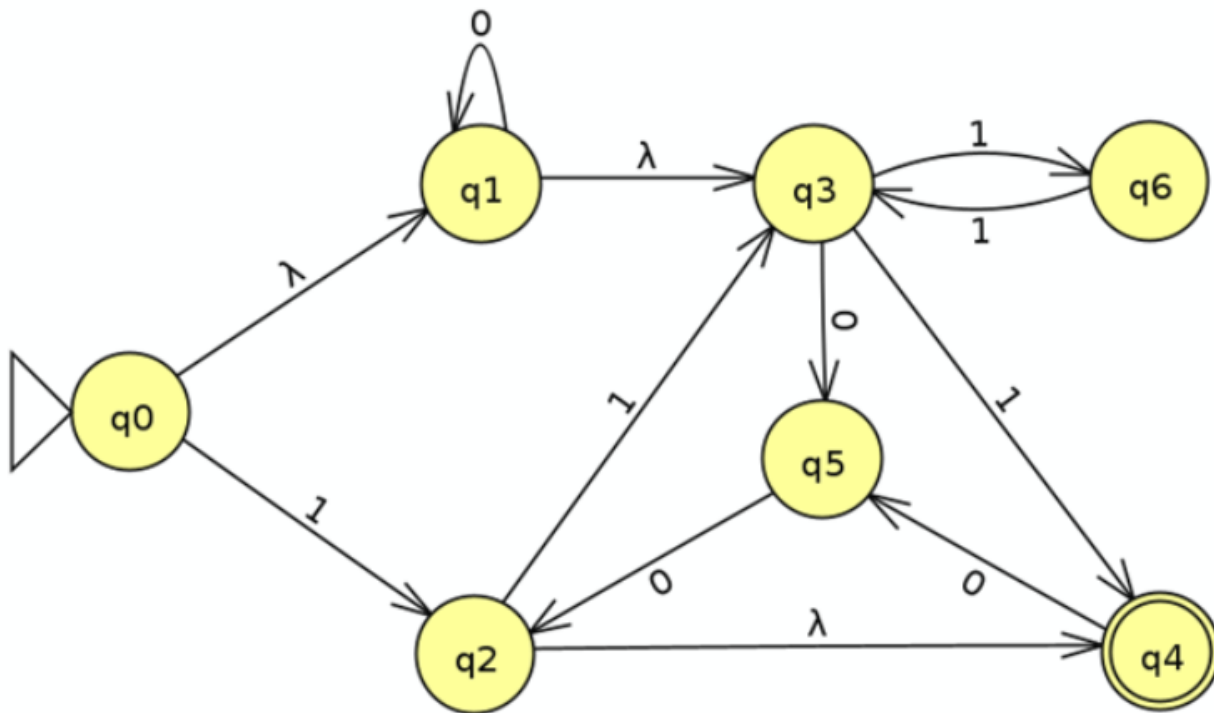


d)

Select the sentences that better describe the language represented by this ϵ -NFA:

"Strings over the alphabet $\{0,1\}$ with a prefix of an arbitrary number of zeros or '11', and the remainder is composed by..." 

(0/1 Ponto)



- ☐ any number of 1's
- ☐ an odd number of 1's
- ☐ the 1's can be mixed with a maximum of two contiguous 0's
- ☐ the 1's can be mixed with an even number of 0's
- ☐ and always ends with a 1